SYSTEM POWER ARCHITECTURE & TOPOLOGICAL SPECIFICATION

AIoT Electric Motorcycle Black Box Accident Recording Device

1. Executive Overview & Design Objectives

This document specifies the complete, buildable, and technically defensible power supply architecture for the AIoT Electric Motorcycle Black Box project. The system functions as a high-reliability accident recording unit deployed on an electric motorcycle. Operating in harsh automotive conditions, the device records critical telemetry (ESP32-S3 processing, MicroSD logging, MPU6050 6-axis IMU, DS3231 RTC, NEO-M8N GNSS) and transmits crash telemetry over 4G LTE cellular network (A7682S).

Reliability and safety take strict precedence over minimum component cost. The architecture is engineered to guarantee continuous, fail-safe operation during extreme transients, sudden vehicle power collapse during a crash, and severe 2.0A cellular RF transmission bursts.

2. System Power Source Relationship & Hierarchy

[CRIT] CRITICAL POWER HIERARCHY RULE
PRIMARY OPERATING POWER SOURCE: 1S 18650 Li-ion Battery (VBAT). EXTERNAL POWER (USB-C / VEHICLE 5V): External charging and supplemental source, NOT the primary operating rail. ESP32 DevKit is a LOAD of the power supply, NOT a power distributor.

The project enforces the official power relationship:

External 5V Inputs (Vehicle 5V_IN / USB 5V_IN) -> TPS2121 Power Mux -> BQ25895 NVDC Charger -> 18650 Battery (VBAT) -> System Rail (SYS) -> TPS61088 Synchronous Boost -> 5V_SYS Main System Rail -> TPS62162 Synchronous Buck -> 3V3_SYS Logic Rail.

Dedicated Cellular Supply: 5V_SYS -> Dual Selection (Option A: 5V P-MOSFET Switch vs Option B: TPS62130 3.8V Buck) -> Selection Jumper JP1 -> 1000uF Low-ESR Polymer Capacitor -> SIM_PWR -> A7682S Breakout.

3. Separated System Power Domains

Power Rail	Voltage Range	Current Rating	Functional Scope & Loads
VBUS_MUX	4.4V - 5.5V	3.0A max	Output of TPS2121 Power Mux; supplies BQ25895
VBAT	3.0V - 4.2V	6.0A trip	1S Li-ion Battery rail; protected by DW01A +
SYS	3.0V - 4.2V	5.0A peak	NVDC System rail from BQ25895; supplies TPS61
5V_SYS	5.00V ±1.5%	3.5A / 5A	Regulated 5V rail; supplies ESP32-S3, 5V Buzz
3V3_SYS	3.30V ±1.5%	1.0A cont	Regulated 3.3V peripheral rail; supplies MPU6
SIM_PWR	5.0V / 3.8V	2.0A burst	Dedicated cellular rail; buffered by 1000uF p
GND	0.0V Return	Common	Unified low-impedance ground plane with star

4. Grounding & Layout Strategy

Due to high di/dt switching transitions (600kHz boost, 2.25MHz buck) and 2.0A periodic cellular transmission pulses (577us pulse width every 4.615ms in 2G/4G mode), the layout enforces strict separation between Power Ground (PGND) and Analog Ground (AGND).

PGND carries heavy circulating currents from inductors L1, L2, L3, L4, input bypass caps, and polymer capacitor C35. AGND carries sensitive analog references (ADC_VBAT_SENSE divider, BQ25895 TS NTC sensor, TPS61088 FB/COMP networks). AGND is tied to PGND at a single star point on the bottom layer directly beneath the BQ25895 thermal pad.

5. Power Path Operating States & Transition Verification

Operating Scenario	System Behavior & Electrical Safety Verification
Case 1: Battery Only	No external power; BQ25895 BATFET is fully ON
Case 2: Vehicle Running	Vehicle 5V present; TPS2121 selects Vehicle i
Case 3: USB-C Plugged	Bench testing / charging; TPS2121 selects USB
Case 4: Dual Input Present	Vehicle 5V and USB-C 5V both present. TPS2121
Case 5: Sudden Vehicle Loss	Vehicle power disappears instantly during acc
Case 6: Battery Discharged	Battery at 3.0V cutoff; A7682S triggers 2A tr
Case 7: SD Write + 4G Burst	MicroSD writes incident sector while 4G trans
Case 8: Battery Absent	18650 cell disconnected or deeply discharged.